Microprocessors And Microcontrollers

Assignment 2- Week 2

TYPE OF QUESTION: MCQ

Number of questions: 15 **Total mark: 15 X 1 = 15**

QUESTION 1:

Correct Answer: b

Detailed Solution: A microprocessor is the central processing unit (CPU) of a computer system and is responsible for executing instructions. It typically contains:

- ALU (Arithmetic Logic Unit): Performs arithmetic and logical operations.
- Registers: Small, fast storage locations that temporarily hold data and instructions.
- Control Unit: Directs the operation of the processor by interpreting instructions and controlling the flow of data between the CPU and other components.

QUESTION 2:

Correct Answer: c

Detailed Solution: 1 MB (Megabyte) is equivalent to 1024 Kilobytes (KB). Here's the breakdown:

- 1 KB = 1024 bytes
- 1 MB = 1024 KB = 1024 * 1024 bytes = 1,048,576 bytes

QUESTION 3:

Correct Answer: c

Detailed Solution: The **Intel 8085 microprocessor** is a **40-pin** integrated circuit (IC). These pins include connections for data, address, power supply, control signals, and interrupt lines. The 8085 microprocessor is an 8-bit processor that was widely used in early microprocessor-based systems.

QUESTION 4:

Correct Answer: b

Detailed Solution: The **program counter (PC)** in the **8085 microprocessor** is a **16-bit register** because it is used to store the address of the next instruction to be executed. The 8085 microprocessor has **16 address lines**, which means it can access **2^16 = 65,536** memory locations (addresses), and hence the program counter needs to be a 16-bit register to store these addresses.

QUESTION 5:

Correct Answer: d

Detailed Solution: The **ALE (Address Latch Enable)** signal in the **Intel 8085 microprocessor** is used to demarcate the time when the address is being sent from the microprocessor to the memory or I/O device. When ALE is **high**, the **lower 8 bits** of the address are placed on the data bus. This allows the data bus to be used as the **low-order address bus** for accessing memory locations.

QUESTION 6:

Correct Answer: b

Detailed Solution: The **flag register** in the **Intel 8085 microprocessor** contains the status flags, which are used to indicate the result of arithmetic and logical operations. These flags are:

- **S (Sign flag)**: Set if the result of an operation is negative.
- **Z (Zero flag)**: Set if the result of an operation is zero.
- **AC (Auxiliary carry flag)**: Set if there is a carry from the lower nibble (used for BCD arithmetic).
- **P (Parity flag)**: Set if the result has an even number of 1's (indicating even parity).
- **CY (Carry flag)**: Set if there is a carry out of the most significant bit (for addition) or a borrow (for subtraction).

These flags are important for conditionally branching or for checking the results of operations.

QUESTION 7:

Correct Answer: d

Detailed Solution: The **Intel 8085 microprocessor** typically operates with a clock speed of **3.125 MHz**. This clock speed is derived from the clock cycle and is the frequency at which the microprocessor's internal operations, such as fetching instructions and executing them, occur.

QUESTION 8:

Correct Answer: a

Detailed Solution: The **flag register** holds the status flags that indicate the result of arithmetic and logical operations. These flags include the **Sign (S)**, **Zero (Z)**, **Auxiliary Carry (AC)**, **Parity (P)**, and **Carry (CY)** flags. These flags provide information about the nature of the result, such as whether the result is zero, negative, or if there was a carry or borrow during the operation.

QUESTION 9:

Correct Answer: b

Detailed Solution: The **CMA (Complement Accumulator)** instruction in the **8085 microprocessor** performs a **1's complement** of the contents of the accumulator. This means that it flips all the bits of the accumulator (i.e., converts 0s to 1s and 1s to 0s).

QUESTION 10:

Correct Answer: c

Detailed Solution: The **INX BC** instruction in the **8085 microprocessor** increments the **16-bit register pair BC** by one. It treats the contents of **B** and **C** together as a 16-bit value and adds one to it. This operation affects both registers (B and C), where the contents of **C** are incremented first, and if there is a carry, the contents of **B** are also incremented.

QUESTION 11:

Correct Answer: b

Detailed Solution: T The **parity flag** in the **8085 microprocessor** is set based on the number of 1s in the result. If the number of 1s in the result is **even**, the parity flag is **set** (indicating even parity). If the number of 1s is **odd**, the parity flag is **reset** (indicating odd parity).

In your case, the register contains **4EH** (which is **01001110** in binary). Let's count the number of 1s:

- **01001110** has **4 ones** (which is an even number).

Since the number of 1s is even, the **parity flag** is **set** (even parity).

QUESTION 12:

Correct Answer: c

Detailed Solution: A **program that uses mnemonics** refers to a **program written in assembly language**. In assembly language, human-readable mnemonics (such as `MOV`, `ADD`, `SUB`, etc.) are used to represent machine-level instructions. These mnemonics are later translated into machine code by an assembler.

QUESTION 13:

Correct Answer: d

Detailed Solution:

1. **RAL (Rotate Accumulator Left)**:
   - The **RAL** instruction performs a left rotation of the contents of the accumulator. This means that the most significant bit (MSB) is shifted into the carry flag, and the contents of the accumulator are shifted left by one position.
   - For **FFH** (which is `11111111` in binary), rotating it left results in **FEH** (`11111110`), and the **carry flag** will be set to 1.
2. **RLC (Rotate Left through Carry)**:
   - The **RLC** instruction performs a left rotation through the carry flag. The most significant bit (MSB) of the accumulator is shifted into the carry flag, and the contents of the accumulator are shifted left by one position, with the carry flag being inserted into the least significant bit (LSB).
   - After the **RAL** instruction, the **carry flag** is 1 (since the MSB of **FFH** is moved into it). When **RLC** is executed, the carry flag (which is now 1) will be placed into the LSB of the accumulator, and the new accumulator content will be **FFH** (`11111111`).

Summary:

- After **RAL**, the accumulator becomes **FEH** (`11111110`).
- After **RLC**, the accumulator becomes **FFH** (`11111111`).

QUESTION 14:

Correct Answer: c

Detailed Solution:

In the **8085 microprocessor**, the **LDAX** instruction is used to load the accumulator with the contents of memory pointed to by a register pair. However, the **LDAX** instruction only works with the **BC** and **DE** register pairs. It loads the accumulator with the value stored at the memory address specified by these register pairs.

- **LDAX BC**: Loads the accumulator with the 16-bit memory address contained in **BC**.
- **LDAX DE**: Loads the accumulator with the 16-bit memory address contained in **DE**.

The **HL** register pair is used for other operations like **LHLD** and **SHLD**, but it is **not** used in the **LDAX** instruction.

QUESTION 15:

Correct Answer: d

Detailed Solution: The **XRA A** instruction in the **8085 microprocessor** performs an **exclusive OR (XOR)** operation between the contents of the accumulator (**A**) and itself.

- **XOR operation** between any number and itself results in **0**.
  - So, **A XOR A = 00** (since any value XORed with itself gives 0).
- **Zero Flag (Z)**:
  - The **Zero Flag** is set (Z = 1) if the result of the operation is **zero**. Since the result of **XRA A** is **00**, the **Zero Flag** will be set to **1**.
- **Carry Flag (CY)**:
  - The **Carry Flag** is not affected by the **XRA** operation, so it remains unchanged. Since the **Carry Flag** was not set before the instruction and there is no carry in this operation, **CY = 0**.

Summary:

After executing **XRA A**:

- **A = 00** (the contents of the accumulator are zero).
- **CY = 0** (the carry flag is not affected).
- **Z = 1** (the zero flag is set because the result is zero).